

# Numerical PDE Quiz 3

May 12, 2026

1. Consider the linear parabolic equation:

$$u_t = \kappa u_{xx} + g(x, t), \quad x \in (0, L), \quad t > 0$$

with B.C.

$$u(0, t) = 0, \quad u_x(L, t) = 0;$$

I.C.

$$u(x, 0) = u_0(x) > 0.$$

Use Crank-Nicolson method for time discretization and the central difference for space discretization.

- (a) Using the ghost point technique at  $x = L$ , write down the fully discrete scheme in component-wise form. Let  $h = \frac{L}{M}$  and  $\tau$  be time step.
  - (b) Using the discrete  $L^2$  norm, show that the scheme is unconditionally stable for  $g = 0$ .
2. Consider following heat equation on  $(0, 2\pi)^d$  with periodic boundary condition:

$$\frac{\partial u}{\partial t} = c^2 \sum_{k=1}^d \frac{\partial^2 u}{\partial x_k^2}$$

- (a) Present the procedures of solving the equation by the forward Euler method.
- (b) Under what condition will FE preserve  $\|U^{n+1}\|_\infty \leq \|U^n\|_\infty$  ?
- (c) For  $d = 3$ , present the basic procedures of the fast solver to solve the linear system obtained by backward Euler methods with help of `fft3` and `ifft3`.
- (d) For any dimension  $d$ , show that the Crank-Nicolson scheme is unconditionally stable by Von Neumann stability analysis.